

Exercise: Using the fact that the angular momentum of a body, about its centre of mass, is

$$\mathbf{H}_G = \int (\mathbf{r} \times \mathbf{v}) dm,$$

where $\mathbf{r}$ is the distance of a piece of mass dm from the centre of mass, and $\mathbf{v} = \boldsymbol{\omega} \times \mathbf{r}$ is its axial velocity, show that the Inertia tensor is

$$\mathbf{I}(\boldsymbol{\omega}) = \begin{bmatrix} \int (y^2 + z^2) dm & -\int xy dm & -\int xz dm \\ -\int xy dm & \int (x^2 + z^2) dm & -\int yz dm \\ -\int xz dm & -\int yz dm & \int (x^2 + y^2) dm \end{bmatrix} \boldsymbol{\omega}$$

The identity

$$\mathbf{A} \times (\mathbf{B} \times \mathbf{C}) = (\mathbf{A} \cdot \mathbf{C})\mathbf{B} - (\mathbf{A} \cdot \mathbf{B})\mathbf{C}$$

will likely prove useful.

$$d\mathbf{H}_G = (\mathbf{r} \times \mathbf{v}) dm = [\mathbf{r} \times (\boldsymbol{\omega} \times \mathbf{r})] dm$$

$$= [(\mathbf{r} \cdot \mathbf{r})\boldsymbol{\omega} - (\mathbf{r} \cdot \boldsymbol{\omega})\mathbf{r}] dm$$

$$= \left[(x^2 + y^2 + z^2) \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} - (x\omega_x + y\omega_y + z\omega_z) \begin{bmatrix} x \\ y \\ z \end{bmatrix} \right] dm$$

$$= \begin{bmatrix} (x^2 + y^2 + z^2)\omega_x - (x\omega_x + y\omega_y + z\omega_z)x \\ (x^2 + y^2 + z^2)\omega_y - (x\omega_x + y\omega_y + z\omega_z)y \\ (x^2 + y^2 + z^2)\omega_z - (x\omega_x + y\omega_y + z\omega_z)z \end{bmatrix}$$

$$= \begin{bmatrix} (y^2 + z^2) & -xy & -xz \\ -xy & (x^2 + z^2) & -yz \\ -xz & -yz & (x^2 + y^2) \end{bmatrix} \begin{bmatrix} \omega_x \\ \omega_y \\ \omega_z \end{bmatrix} dm$$

$$\therefore H_G = I(\omega)$$

$$= \begin{bmatrix} \int y^2 + z^2 dm & -\int xy dm & -\int xz dm \\ -\int xy dm & \int x^2 + z^2 dm & -\int yz dm \\ -\int xz dm & -\int yz dm & \int x^2 + y^2 dm \end{bmatrix} \omega$$

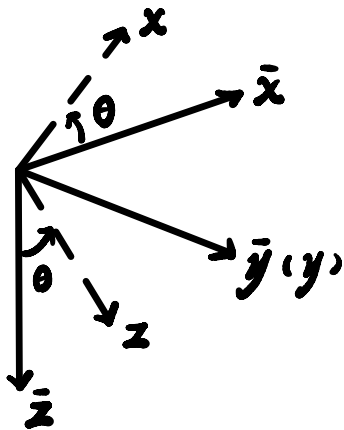
Exercise:

Consider the inertial properties of a given aircraft for a given level of fuel mass to have been provided using a certain set of body axes, with coordinates $\bar{x}$, $\bar{y}$, $\bar{z}$, such that the inertia tensor has component form

$$\mathbf{I} = \begin{bmatrix} I_{\bar{x}\bar{x}} & -I_{\bar{x}\bar{y}} & -I_{\bar{x}\bar{z}} \\ -I_{\bar{x}\bar{y}} & I_{\bar{y}\bar{y}} & -I_{\bar{y}\bar{z}} \\ -I_{\bar{x}\bar{z}} & -I_{\bar{y}\bar{z}} & I_{\bar{z}\bar{z}} \end{bmatrix}.$$

and the $\bar{x}\bar{z}$ -plane is the vehicle's plane of symmetry. When performing flight dynamics analyses you will often have to re-estimate the inertia tensor for a different, more convenient set of wind axes.

- Calculate the component form of the same inertia tensor in new set of body axes (with coordinates x , y , z) obtained from the old by rotating the body about the $\bar{y}$ axis by an angle θ . To do so, first calculate the new components by direct computation. Then check your results by exploiting the fact that the inertia tensor is an object which exists independently of any coordinate system.
- Use your findings to compute the orientation of the principal axes of the aircraft with respect to the old axes, if we can assume the aircraft to be laterally symmetric.
- How would you adjust the vehicle's inertial tensor to account for a change of its onboard fuel mass Δm_F , if the fuel tanks' centre of gravity is at $\bar{x}_F$, $\bar{y}_F$, $\bar{z}_F$. You may assume the fuel tanks may be treated as point masses and fuel is consumed evenly about the aircraft's axis of symmetry.



$$x = \bar{x} \cos \theta + \bar{z} \sin \theta$$

$$y = \bar{y}$$

$$z = \bar{z} \cos \theta - \bar{x} \sin \theta$$

$$I_{xx} = \int (y^2 + z^2) dm$$

$$= \int (\bar{y}^2 + \bar{z}^2 \cos^2 \theta + \bar{x}^2 \sin^2 \theta - 2\bar{x}\bar{z} \cos \theta \sin \theta) dm$$

$$= I_{\bar{x}\bar{x}} \cos^2 \theta + I_{\bar{z}\bar{z}} \sin^2 \theta - 2I_{\bar{x}\bar{z}} \sin \theta \cos \theta$$

$$I_{xy} = I_{yx} = \int xy dm = I_{\bar{x}\bar{y}} \cos \theta + I_{\bar{z}\bar{y}} \sin \theta$$

$$I_{yy} = I_{\bar{y}\bar{y}}$$

$$I_{yz} = I_{\bar{y}\bar{z}} \cos \theta - I_{\bar{x}\bar{y}} \sin \theta$$

$$I_{xz} = \int \bar{x}\bar{z} \cos^2 \theta - \bar{x}^2 \sin \theta \cos \theta + \bar{z}^2 \sin \theta \cos \theta - \bar{x}\bar{z} \sin^2 \theta dm$$

$$= I_{\bar{x}\bar{z}} (\cos^2 \theta - \sin^2 \theta) + (I_{\bar{x}\bar{x}} - I_{\bar{z}\bar{z}}) \sin \theta \cos \theta$$

$$I_{zz} = I_{\bar{z}\bar{z}} \cos^2 \theta + I_{\bar{x}\bar{x}} \sin^2 \theta - 2I_{\bar{x}\bar{z}} \sin \theta \cos \theta$$

Let $\mathbf{v} = \mathbf{I}(\mathbf{u})$, thus, $\bar{\mathbf{v}} = \bar{\mathbf{I}}(\mathbf{u})$:

$$\begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} \bar{v}_1 \\ \bar{v}_2 \\ \bar{v}_3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} \bar{v}_1 \\ \bar{v}_2 \\ \bar{v}_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix}^{-1} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$\begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix} = \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix}^{-1} \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$= \begin{bmatrix} \cos\theta & 0 & \sin\theta \\ 0 & 1 & 0 \\ -\sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \begin{bmatrix} \cos\theta & 0 & -\sin\theta \\ 0 & 1 & 0 \\ \sin\theta & 0 & \cos\theta \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$\therefore \begin{bmatrix} v_1 \\ v_2 \\ v_3 \end{bmatrix}$$

$$= \begin{bmatrix} I_{xx} \cos \theta - I_{xz} \sin \theta & -I_{xy} \cos \theta - I_{yz} \sin \theta & -I_{xz} \cos \theta + I_{xx} \sin \theta \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} \sin \theta - I_{zx} \cos \theta & I_{yz} \sin \theta - I_{zy} \cos \theta & I_{xz} \sin \theta + I_{zx} \cos \theta \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & -\sin \theta \\ 0 & 1 & 0 \\ \sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}$$

$$= \begin{bmatrix} I_{xx} & -I_{xy} & -I_{xz} \\ -I_{xy} & I_{yy} & -I_{yz} \\ -I_{xz} & -I_{yz} & I_{zz} \end{bmatrix} \begin{bmatrix} \bar{u}_1 \\ \bar{u}_2 \\ \bar{u}_3 \end{bmatrix}$$

$$I_{xz} = I_{xz} (\cos^2 \theta - \sin^2 \theta) + (I_{xx} - I_{zz}) \sin \theta \cos \theta = 0$$

$$\therefore I_{xz} \cos(2\theta) + \frac{1}{2} (I_{xx} - I_{zz}) \sin(2\theta) = 0$$

$$\therefore \theta = \frac{1}{2} \tan^{-1} \left(\frac{2I_{xz}}{I_{zz} - I_{xx}} \right)$$

$$I_o = \begin{bmatrix} I_{xx} & 0 & -I_{xz} \\ 0 & I_{yy} & 0 \\ -I_{xz} & 0 & I_{zz} \end{bmatrix} \quad \text{where } y = 0$$

$$\therefore y' = 0 \text{ for lateral symmetric} \quad \therefore \Delta y = 0$$

Take moment about origin:

$$mX = (m - m_F) X' + m_F X_F \quad \therefore \Delta X = X' - X = \frac{m_F (X - X_F)}{m - m_F}$$

$$mZ = (m - m_F) Z' + m_F Z_F \quad \therefore \Delta Z = Z' - Z = \frac{m_F (Z - Z_F)}{m - m_F}$$

$$\therefore I_{xx}' = \sum [(y_i - Y)^2 + (z_i - Z)^2] m_i$$

$$\therefore I_{xx}' = I_{xx} - (m - m_F) \Delta Z^2 - m_F (Z - Z_F)^2 - m_F y_F^2$$

$$I_{zz}' = I_{zz} - (m - m_F) \Delta X^2 - m_F (X - X_F)^2 - m_F y_F^2$$

$$I_{yy}' = I_{yy} - (m - m_F) \Delta Z^2 - m_F (Z - Z_F)^2 - (m - m_F) \Delta X^2 - m_F (X - X_F)^2$$

$$I_{xy}' = I_{yx}' = 0$$

$$I_{xz}' = I_{zx}' = I_{xz} - (m - m_F) \Delta X \Delta Z - m_F (X - X_F)(Z - Z_F)$$

Exercise:

You should remember that, although not immediately apparent from the formulation above, further non-linearity can enter these equations in the form of the forces and moments acting on the vehicle, which often themselves depend in a non-linear way on the various degrees of freedom of the aircraft. For example in figure 8 you can see the effect of introducing wing-strakes in an attempt to produce a more linear relationship between lift force, rolling moment and the aircraft's angle of attack, and hence to improve the handling qualities of the vehicle at high angles of attack.

- What other cases of non-linearity in the vehicle's aerodynamics can you think of?

see answers ...

Exercise:

Wing In Ground Effect vehicles, often referred to as WIGs or Ekranoplans, operate at heights that are a fraction of their wing chord or wing span. Operating that close to the surface greatly affects the vehicle's aerodynamics and therefore to study their dynamics, the vehicle's height above the surface h cannot be neglected from the analysis.

Assuming wings-level flight, derive the kinematic equation of motion that relates the vehicle's height to the vehicle's velocities and pitch attitude, as defined with respect to its body-fixed axis system.

$$\begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix} = R_{EB}(\Phi, \Theta, \Psi) \begin{bmatrix} U \\ V \\ W \end{bmatrix} = R_{E1}(\Psi) R_{12}(\Theta) R_{23}(\Phi) \begin{bmatrix} U \\ V \\ W \end{bmatrix}$$

$$\therefore \Phi = 0 \quad \therefore R_{23}(\Phi) = I$$

$$\therefore \begin{bmatrix} \dot{x}_E \\ \dot{y}_E \\ \dot{z}_E \end{bmatrix} = \begin{bmatrix} \cos \varphi & -\sin \varphi & 0 \\ \sin \varphi & \cos \varphi & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} \cos \theta & 0 & \sin \theta \\ 0 & 1 & 0 \\ -\sin \theta & 0 & \cos \theta \end{bmatrix} \begin{bmatrix} U \\ V \\ W \end{bmatrix} \quad * h = -z_E$$

Exercise:

Carry out the analysis for the mass-spring-damper system discussed in the prior example for a non-zero damping coefficient c .

- What structure of trajectories emerges for $c^2 + 4k > 0$?
- What structure of trajectories emerges for $c^2 + 4k = 0$?
- What structure of trajectories emerges for $c^2 + 4k < 0$?

Plot the trajectories near the singular point for a range of numerical values for c and k . What changes in the parameters mark the difference between a 'trivial' change in structure and a 'significant' change in structure?

$$\lambda = \frac{c \pm \sqrt{c^2 + 4k}}{2} \quad \text{. singular point } (0, 0)$$

$$\textcircled{1} \quad c^2 + 4k = 0$$

$$c^2 = -4k \longrightarrow \lambda = \frac{c}{2} \begin{cases} c > 0 \longrightarrow \text{source} \\ c < 0 \longrightarrow \text{sink} \end{cases}$$

$$\textcircled{2} \quad c^2 + 4k < 0$$

$$\lambda = \frac{c}{2} \pm i \frac{\sqrt{-(c^2 + 4k)}}{2} \begin{cases} c > 0 \longrightarrow \text{unbounded} \\ c < 0 \longrightarrow \text{bounded} \end{cases}$$

$$\textcircled{3} \quad c^2 + 4k > 0$$

$$\lambda = \frac{c}{2} \pm \frac{\sqrt{c^2 + 4k}}{2} \quad \dots$$

Real World Examples:

Several fighter aircraft suffer from wing-rock and/or nose-slice at high angles of attack. Since the 1970's several studies have therefore been conducted to better understand these effects. Those interested should review the papers by Ericsson [10] and Greer [9] and attempt to answer the following questions.

- Can you name any of them?
- How does the pilot experience the dynamic departure associated with nose slice?
- How might a pilot recover the aircraft from a nose-slice or the wing-rock regime?
- What features in the aerodynamics of the aircraft affect the onset of wing-rock and nose-slice?
- How can the aircraft's tendency to exhibit such behaviours be addressed in the design of the aircraft?

- *nose-slice : pitchfork bifurcation*
- wing-rock : Supercritical Hopf bifurcation*
- *end-up in a highly yamed state*
- *reduce the elevator deflection (AoA)*
- *form of stabilization : large rudder ...*
- * *detail see Dutch-roll, where nose-slice + wing-rock*

Exercise:

In the introduction section we made mention of the tricky handling characteristics of the Sopwith Camel. Designed to be a light, compact and agile fighter, it was powered by a 150 hp Bentley BR1 rotary engine, weighting 400 lbs and rotating at 1250 rpm. Novice pilots often had trouble contriling the aircraft as pitching control inputs often resulted in the aircraft also responding in yaw. One instructor at the Central Flying School at Upavon, Wiltshire, wrote “*In spite of all the care we took, Camels continually spun down out of control when flown by pupils on their first solos.*” [14]. This is due to the relatively large gyroscopic forcing generated by the Camel’s rotary engine.

In vector form, the gyroscopic moment generated by an axisymmetric body rotating with an angular velocity $\omega_B = [\omega_x, \omega_y, \omega_z]$, in response to an external rotational pertur-

bation $[p, q, r]$, will be

$$\begin{bmatrix} L \\ M \\ N \end{bmatrix} = \begin{bmatrix} I_{xx_B} \omega_x \\ I_{yy_B} \omega_y \\ I_{zz_B} \omega_z \end{bmatrix} \times \begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} (2I_{xz_B} - I_{yz_B})q \\ (3I_{yz_B} - I_{xz_B})r \\ (1I_{yz_B} - 2I_{xz_B})p \end{bmatrix}$$

Note that the rotating body's inertial characteristics and rotational velocity are defined with respect to the aircraft's body-fixed system of axes.

Can the gyroscopic effect of a rotating axisymmetric body onboard an aircraft be incorporated into our equations of motion for wings level steady flight? Derive the linearized equations of motion in the case of a rotary engine, whose axis of rotation lies on the aircraft's plane of symmetry and at an angle $\epsilon + \alpha_e$ to the x body-fixed axis. How would that affect any assumptions or simplifications made thus far?

$$\mathbf{x} = \begin{bmatrix} X \\ W \\ q \\ \Theta \\ Y \\ p \\ r \\ \Phi \\ \Psi \end{bmatrix}$$

$$\therefore \mathbf{M}\dot{\mathbf{x}} = \mathbf{N}(\mathbf{x}) + \mathbf{F}(\dot{\mathbf{x}}, \mathbf{x}) + \mathbf{G}(\mathbf{x})$$

At singular point,

$$\mathbf{M}\dot{\mathbf{x}}_0 = 0 = \mathbf{N}(\mathbf{x}_0) + \mathbf{F}(0, \mathbf{x}_0) + \mathbf{G}(\mathbf{x}_0)$$

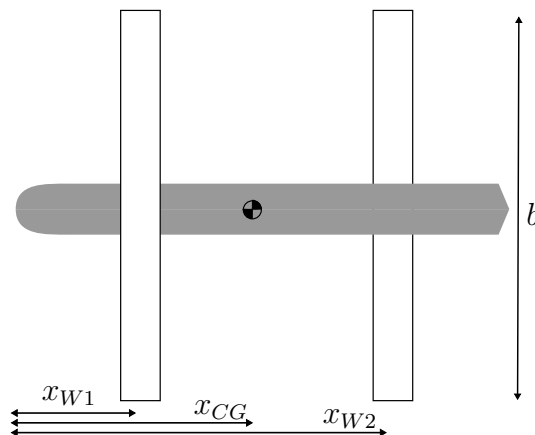
$$\therefore \mathbf{M}\dot{\mathbf{x}}' = (\mathbf{M} - \mathbf{F}_{\dot{\mathbf{x}}_0})\dot{\mathbf{x}}' = (\mathbf{N}_{\mathbf{x}_0} + \mathbf{F}_{\mathbf{x}_0} + \mathbf{G}_{\mathbf{x}_0})\mathbf{x}'$$

$$\mathbf{F} = \begin{bmatrix} X'' \\ Z'' \\ M \\ 0 \\ Y'' \\ L \\ N \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} X'' \\ Z'' \\ M'' + M''^w \\ 0 \\ Y'' \\ L'' + L''^w \\ N'' + N''^w \\ 0 \\ 0 \end{bmatrix}$$

... see answer sheet

Exercise:

Consider a rocket propelled, tandem wing aerial vehicle, a top view of which is shown in the figure below. It features two unswept wings, both having equal planform areas S , aspect ratios $\mathcal{R}$, and lift curve slopes a_W . The forward one's aerodynamic centre is located x_{W1} aft of the datum, and the rear wing's at x_{W2} . The rear wing is affected by the downwash of the front one, having a downwash derivative $d\varepsilon/d\alpha$. The forward wing is affected by the upwash of the rear wing, with an upwash derivative $d\varepsilon_u/d\alpha$.



If the rocket motor's thrust output is independent of variations in the vehicle's velocity and the vehicle's height is minimal relative to its length, derive expressions for this vehicle's longitudinal stability derivatives.

$$\frac{\partial T}{\partial V_{\infty}} \approx 0, \quad Z_{W2} - Z_{CG} \approx 0, \quad Z_{W1} - Z_{CG} \approx 0$$

$$T_x = T \cos(\alpha_e + \epsilon), \quad T_z = T \sin(\alpha_e + \epsilon)$$

$$\text{When } \alpha = \alpha_e, \quad V_{\infty} = U_e$$

$$X = T_x - \frac{1}{2} \rho U_e^2 S_{ref} C_0$$

$$Z = -T_z - \frac{1}{2} \rho U_e^2 S_{ref} C_L$$

$$M = T l_T + \frac{1}{2} \rho U_e^2 S_{ref} \bar{c} (C_{m0} - K_n C_L)$$

$$\begin{aligned} \text{where } l_T &= -(Z_T - Z_{CG}) \cos(\alpha_e + \epsilon) - (X_T - X_{CG}) \sin(\alpha_e + \epsilon) \\ &= (X_{CG} - X_T) \sin(\alpha_e + \epsilon) \quad \text{for this case} \end{aligned}$$

When small perturbation occurs,

$$X = T_x - \frac{1}{2} \rho V_{\infty}^2 S_{ref} [C_0 \cos(\alpha - \alpha_e) - C_L \sin(\alpha - \alpha_e)]$$

$$Z = -T_z - \frac{1}{2} \rho V_{\infty}^2 S_{ref} [C_L \cos(\alpha - \alpha_e) + C_0 \sin(\alpha - \alpha_e)]$$

$$M = T l_T + \frac{1}{2} \rho V_{\infty}^2 S_{ref} \bar{c} (C_{m0} - K_n C_L)$$

$$\therefore \frac{\partial V_{\infty}}{\partial U} = 1, \quad \frac{\partial V_{\infty}}{\partial W} = \frac{\partial V_{\infty}}{\partial M} = 0$$

$$\frac{\partial \alpha}{\partial W} = \frac{1}{U_e}, \quad \frac{\partial \alpha}{\partial U} = \frac{\partial \alpha}{\partial M} = 0$$

$$\therefore X_u^a = \frac{\partial X}{\partial V_\infty} = -\rho U_e S_{ref} (C_D)_e - \frac{1}{2} \rho U_e^2 S_{ref} \left(\frac{\partial C_D}{\partial V_\infty} \right)_e$$

$$Z_u^a = \frac{\partial Z}{\partial V_\infty} = -\rho U_e S_{ref} (C_L)_e - \frac{1}{2} \rho U_e^2 S_{ref} \left(\frac{\partial C_L}{\partial V_\infty} \right)_e$$

$$M_u^a = \frac{\partial M}{\partial V_\infty} = \frac{1}{2} \rho U_e^2 S_{ref} \bar{c} \left[\left(\frac{\partial C_{m0}}{\partial V_\infty} \right)_e - \frac{\partial K_n}{\partial V_\infty} (C_L)_e - K_n \left(\frac{\partial C_L}{\partial V_\infty} \right)_e \right]$$

In addition, $\frac{\partial}{\partial V_\infty} = \frac{1}{C_s} \frac{\partial}{\partial M_\infty}$

$$\therefore X_w^a = \frac{1}{U_e} \frac{\partial X}{\partial \alpha} = -\frac{1}{2} \rho U_e S_{ref} \left[\left(\frac{\partial C_D}{\partial \alpha} \right)_e - (C_L)_e \right]$$

$$Z_w^a = \frac{1}{U_e} \frac{\partial Z}{\partial \alpha} = -\frac{1}{2} \rho U_e S_{ref} \left[\left(\frac{\partial C_L}{\partial \alpha} \right)_e + (C_D)_e \right]$$

$$M_w^a = \frac{1}{2} \rho U_e S_{ref} \bar{c} \left(-K_n \frac{\partial C_L}{\partial \alpha} \right)_e$$

where $(C_D)_e = C_{D0} + \frac{1}{\pi A Re} (C_L)_e^2$

$$\left(\frac{\partial C_L}{\partial \alpha} \right)_e = a_w \left(1 + \frac{d\epsilon_u}{d\alpha} \right) + a_w \left(1 - \frac{d\epsilon}{d\alpha} \right)$$

$$\left(\frac{\partial C_D}{\partial \alpha} \right)_e = \frac{2}{\pi A Re} \left(\frac{\partial C_L}{\partial \alpha} \right)_e$$

for q , $U_i = -(z_H - z_{cg}) q$ and $W_i = (x_H - x_{cg}) q$

In this case, $W_i = (x_{w1} - x_{cg}) q + (x_{w2} - x_{cg}) q$

$$\therefore W_q = x_{w1} + x_{w2} - 2x_{cg}$$

$$X_q = \frac{\partial x}{\partial w_i} \frac{\partial w_i}{\partial q} = X_w (X_{w_i} + X_{w_s} - 2X_{cg})$$

$$Z_q = \frac{\partial z}{\partial w_i} \frac{\partial w_i}{\partial q} = Z_w (X_{w_i} + X_{w_s} - 2X_{cg})$$

$$M_q = \frac{\partial M}{\partial x} \frac{\partial x}{\partial q} + \frac{\partial M}{\partial z} \frac{\partial z}{\partial q} + \left(\frac{\partial M}{\partial W} \right)_c \frac{\partial W}{\partial q}$$

$$= (X_{w_i} + W_{w_s} - 2X_{cg})(X_q + M_w)$$

Question:**General aviation**

Considering the flight of a GA aircraft and that of a large airliner, what will the likely periods of oscillation due to an excitation of the Phugoid mode be?

What about the time to half-amplitude? How might the low damping of this mode affect the aircraft's handling qualities?

For $M = 0.65$, $U_e = 222.95 \text{ m/s}$

$$\lambda = \pm i\sqrt{2} \frac{g}{U_e}$$

$$\therefore T_d = \sqrt{2}\pi \frac{U_e}{g} = 100.97 \text{ (s)}$$

$$\therefore \zeta \approx \frac{1}{\sqrt{2}(L/D)_e} = 0.04159$$

$\uparrow$
 17 for Boeing 747

Exercise:

The procedures above can be used to find the control derivatives any control surface based on the effect that their deflection has on the aircraft's aerodynamics. Hence derive expressions for the control derivatives of

- an all-moving tailplane, where i_H is a control input.
- a spoiler, where its symmetric deflection angle δ_S is a control input. You may assume that the wing's sectional lift and drag varies linearly with spoiler deflection.
- a thrust vector-able engine, where the thrust vector angle ϵ is a control input.

$$\cdot C_{LH} = \frac{S_H}{S_{ref}} a_H \left[\left(1 - \frac{d\epsilon}{d\alpha}\right) \alpha + i\omega - \alpha_{0H} + i\eta \right]$$

$$C_{DH} = C_{D0} + \frac{1}{\pi A_H e_H} C_{LH}^2$$

$$X_{iH} = -\frac{1}{2} \rho U_\infty^2 S_{ref} \left(\frac{dC_{mH}}{d\delta_{iH}} \right) e$$

$$Z_{iH} = -\frac{1}{2} \rho U_\infty^2 S_{ref} \left(\frac{dC_{mH}}{d\delta_{iH}} \right) e$$

$$M_{iH} = (x_H - x_{cg}) Z_{iH} - (z_H - z_{cg}) X_{iH} + \frac{1}{2} \rho U_\infty^2 S_{ref} \bar{c} \left(\frac{\partial C_{mH}}{\partial \delta_{iH}} \right) e$$

• same as above

• same as trust but with $\frac{d}{d\epsilon}$

Exercise:

Section 11.5 presents an approximate method for estimating the elevator deflection required per unit g loading in a symmetric pull-up manoeuvre. The method presented is somewhat crude and more precise results can be obtained using the transfer function derived above.

- Show that the load factor due to the normal acceleration of a vehicle is given by

$$n_c = -\frac{\dot{W}}{g} + q \frac{U_e}{g}$$

- Hence derive an expression for the steady state value of the elevator deflection required per unit g.
- How does this compare with equation (178) given by the derivation in section 11.5? What might the source of any discrepancy be?

Exercises:

- List as many system outputs, that you could use to control an aircraft based on, as you can think of. What device would you use to measure each of them.
- How valid is the assumption made in the prior example, that the measured airspeed by the aircraft's pitot static system is not affected by the aircraft's pitch rate?
- Estimate the $\mathbf{C}$ matrix if we were to model a measurement of the aircraft's true airspeed and angle of attack and $\mathbf{x} = [U, W, q, \Theta]$.
- Estimate the $\mathbf{C}$ matrix if we were to model a measurement of the aircraft's rate of climb and $\mathbf{x} = [U, W, q, \Theta]$.
- You have a system

$$\mathbf{M}\dot{\mathbf{x}} = \mathbf{R}\mathbf{x} + \mathbf{B}\mathbf{u}$$

$$\mathbf{y} = \mathbf{C}\mathbf{x}$$

with n states, m control inputs and p outputs. If $\mathbf{u} = \mathbf{K}\mathbf{y}$, what would the dimensions of the $\mathbf{M}$, $\mathbf{R}$, $\mathbf{B}$, $\mathbf{C}$ and $\mathbf{K}$ matrices be?

- *see solution*
- *see solution*

$$\cdot \mathbf{C} = \begin{bmatrix} \frac{\partial V_\infty}{\partial U} & \frac{\partial V_\infty}{\partial W} & \frac{\partial V_\infty}{\partial q} & \frac{\partial V_\infty}{\partial \theta} \\ \frac{\partial \alpha}{\partial U} & \frac{\partial \alpha}{\partial W} & \frac{\partial \alpha}{\partial q} & \frac{\partial \alpha}{\partial \theta} \end{bmatrix}_e = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & \frac{1}{U_e} & 0 & 0 \end{bmatrix}$$

$$\cdot \dot{h} = U \sin \theta - W \cos \theta$$

$$\mathbf{C} = \dots$$

$$\cdot \mathbf{M}, \mathbf{R} = n \times n$$

$$\mathbf{B} = n \times m$$

$$\mathbf{C} = p \times n$$

$$\mathbf{K} = m \times p$$

Exercise:

Consider the system

$$m\ddot{x}(t) - c\dot{x}(t) - kx(t) = bu(t)$$

where $c < 0$, $b > 0$,

$$u(t) = K_p(r - x) \quad \text{and} \quad K_p \geq 0$$

- Calculate the system's response to any external perturbation when $r = 0$. Use a root-locus plot to illustrate the effect that the proportional gain K_p has on the system's dynamics.
- Show that when $r \neq 0$, the steady state error is inversely proportional to the magnitude of K_p .

$$u(t) = K_p (r - x)$$

$$\therefore m\ddot{x} - c\dot{x} - kx = bK_p(r - x)$$

$$\therefore \begin{bmatrix} m & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} = \begin{bmatrix} c & k - bK_p \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix} + \begin{bmatrix} bK_p \\ 0 \end{bmatrix} r$$

when $r = 0$,

$$\begin{aligned} \begin{bmatrix} \ddot{x} \\ \dot{x} \end{bmatrix} &= \begin{bmatrix} m & 0 \\ 0 & 1 \end{bmatrix}^{-1} \begin{bmatrix} c & k - bK_p \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix} \\ &= \frac{1}{m} \begin{bmatrix} 1 & 0 \\ 0 & m \end{bmatrix} \begin{bmatrix} c & k - bK_p \\ 1 & 0 \end{bmatrix} \begin{bmatrix} \dot{x} \\ x \end{bmatrix} \end{aligned}$$

for stability,

$$\begin{vmatrix} c - m\lambda & k - bK_p \\ 1 & -\lambda \end{vmatrix} = 0$$

$$m\lambda^2 - c\lambda - k + bK_p = 0 \quad \therefore \lambda = \frac{c \pm \sqrt{c^2 - 4m(bK_p - k)}}{2m}$$